

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. (EC) Examination

Nov 2013

EC210 Analog Communication

Date: 15.11.2013, Friday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q – 1 State and Prove following Fourier Transform Properties: Symmetry, Time Shifting and Frequency Shifting. [07]
- Q - 2 (a) Explain modulation index of AM in detail. [03]
- (b) Determine efficiency η and the percentage of the total power carried by sidebands of AM wave for modulation when index is (i) $\mu=0.5$ and (ii) $\mu=0.3$ [04]
- (c) Explain detection of amplitude modulated signal using diode detector. [07]

OR

- Q - 2 (a) Define: (i) Energy spectral density (ii) Power spectral density [03]
- (b) Find the Fourier transforms of the everlasting sinusoid functions $\cos \omega_0 t$ and $\sin \omega_0 t$ [04]
- (c) Explain SSB pilot carrier radio system. [07]
- Q - 3 (a) Describe Automatic Frequency Control (AFC). [03]
- (b) Explain Ratio Detector circuit. [04]
- (c) Explain FM stereo broadcast transmitter with block diagram in detail. [07]

OR

- Q - 3 (a) What is pre-emphasis and de-emphasis network? [03]
- (b) Explain phase-locked loop FM Demodulator. [04]
- (c) Explain FM broadcast receiver with block diagram in detail. [07]

SECTION – II

- Q - 4 Write short note on: Thermal Noise, Shot Noise, Partition Noise and Burst noise. [07]
- Q - 5 (a) Explain Single Side Band Generation. [07]
- (b) The equivalent noise resistance for an amplifier is 300Ω and the equivalent shot noise current is $5\mu A$. The amplifier is fed from a 150Ω , $10\mu V$ rms sinusoidal signal source. Calculate the individual noise voltages at the input and the input signal to noise ratio in decibel. The noise bandwidth is 10 MHz. [07]

OR

- Q - 5 (a) Explain super heterodyne receiver with neat block diagram. [07]
- (b) Derive Friis's formula for N number of amplifier connected in cascade connection. [07]
- Q - 6 (a) A receiver tunes signals from 550 to 1600 kHz with an IF of 455 kHz. Find the frequency tuning ranges and capacitor tuning ranges for the oscillator section and for the RF section. [07]
- (b) Explain Single Side Band Reception. [07]

OR

- Q - 6 (a) Explain Automatic Gain Control (AGC). [07]
- (b) Compare Amplitude modulation and Frequency modulation. [07]